

Classifying Suicide-Related Communication on Reddit

Ruby Villalona – Montclair State University, villalona1@montclair.edu

Subtle linguistic signs of suicidal ideation are present in postings on Reddit. It is impossible to check all the content by moderators that would result in high-risk posts being overlooked. The study examines language bias on NLP suicide-risk classification based on posts that have been labeled as *Supportive*, *Ideation*, *Behavior* or *Attempt* and bias metrics (*bias mean*, *bias TF-IDF*).

Online language can be used to determine suicide risk. This was framed in CLPsych 2019 Shared Task as a multi-level classification problem with Supportive and Ideation posts being easier to classify and Behavior and Attempt overlap confusing (Zirikly et al., 2019). Past literature concentrates on precision not representational bias which is precisely analyzed in this project

The dataset consists of 500 anonymized posts on Reddit, one post per user, and four categories. Posts are brief, informal and have actual statements of distress. Linguistic cues are conserved with minimal cleaning. TF-IDF vectors are passed through a multiclass Logistic Regression model.

As Figure 1 illustrates, Ideation and Supportive content are the least and most biased, respectively, whereas Behavior and Attempt are in the middle range of the spectrum. the most dangerous Attempt posts are less biased, indicating that immediate-risk language is less emotionally differentiated. Figure 2 Class distribution of suicide-related Reddit posts, showing that Ideation posts are most common while Attempt posts are least frequent. Figure 3 (TF-IDF bias scores) has a near to zero normal distribution, indicating that the TF-IDF weighting partially balances affective signals thus decreasing bias.

These results suggest that existing NLP models might overweight ideation language and undervalue real behavioral risk, which can lead to the classification of high-risk posts as false. The controversy between the mean-based and TF-IDF representations represents the conflict between the discriminatory ability and bias reduction. To detect suicide risk accurately, bias-aware feature presentation and consideration of high-risk classes over overall accuracy are necessary to eliminate the critical failures in crisis detection

Appendix

Figure 1 shows Confusion matrix of TF-IDF classifier described in paragraph 4.

```
Confusion Matrix (rows=true, cols=pred):
      pred_Attempt  pred_Behavior  pred_Ideation  pred_Indicator  \
true_Attempt      0           3           3           2
true_Behavior      1           5           6           1
true_Ideation      0           6          11           9
true_Supportive    0           0           5           8
```

Attempt: attempt, somebody, suicide attempt, failed
Behavior: cutting, hospital, felt, kill im
Ideation: thank, depression, mental depression, mental
Supportive: think, sounds, therapist help, chocolate

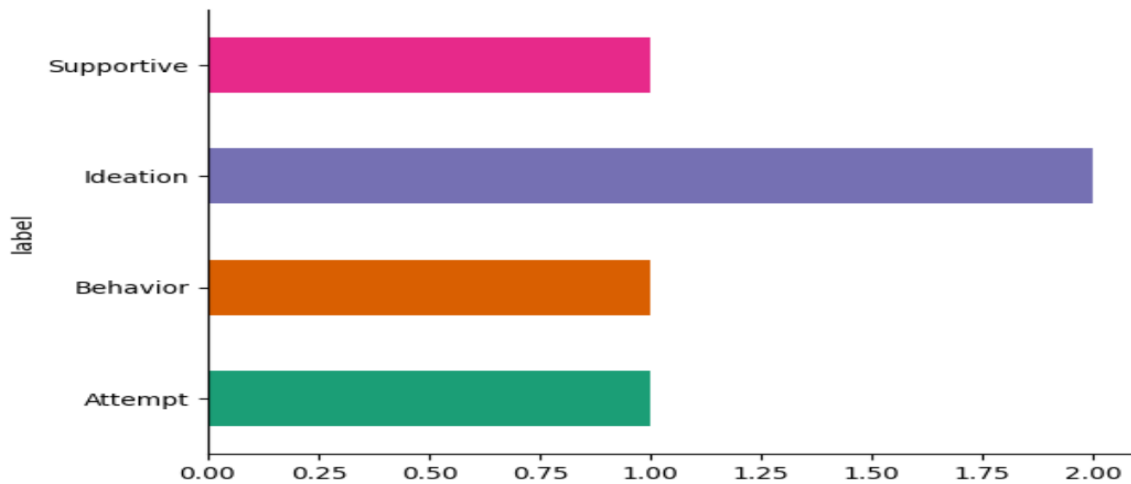
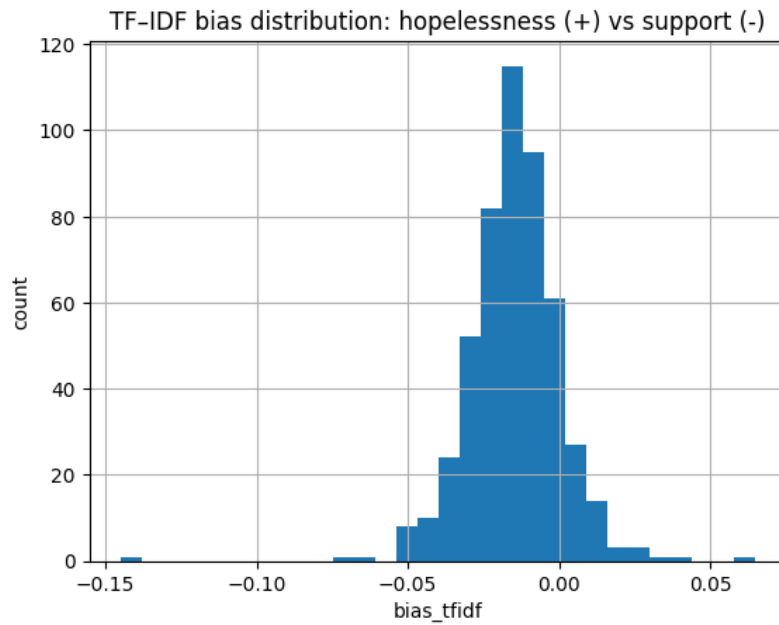


Figure 2 shows Class distribution of suicide-related Reddit posts described in paragraph



4.

Figure 3 shows a Histogram of TF-IDF Bias Scores in Hopelessness-Support Classification described in paragraph 4.

Reference

- Zirikly, A., Resnik, P., Uzuner, Ö., & Hollingshead, K. (2019). *CLPsych 2019 Shared Task: Predicting the degree of suicide risk in Reddit posts*. Proceedings of the Sixth Workshop on Computational Linguistics and Clinical Psychology, 1–12.
- Reddit mental-health posts dataset from Kaggle — *Text Summary with Reddit Mental Health Posts* (<https://www.kaggle.com/code/shriyapr/text-summary-with-reddit-mental-health-posts/input>).